

Light and Photosynthesis

Student worksheet | Biology 2e, Chapter 8 | 45-60 minutes

Name: _____ Class: _____ Date: _____

1. Predict and plan

How does light level affect the time it takes leaf disks to float?

First replace trapped air in leaf disks with solution so they sink. Then use the lamp to explore how light changes oxygen accumulation and flotation. For a controlled comparison, keep each trial at its assigned light level for 20 minutes; you can also change the lamp during a trial and examine the exploratory results.

Prediction and scientific reasoning:

Variables and controls, or sampling plan:

Keep consistent: 10 same-sized disks, same leaf type, 0.2% sodium bicarbonate solution with a trace of detergent, solution depth, and temperature. Only light level changes.

Select your route

Part	Virtual / Classroom	Equipment or conditions
Dark control		
Low light		
Bright light		

Use the same questions for either route. Always label the source of each dataset.

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2. Collect observations

Record floating disks (of 10). Repeat each treatment with fresh material for three independent trials.

Treatment / trial	0 min	2 min	4 min	6 min	8 min	10 min	12 min	14 min	16 min	18 min	20 min
Dark control / 1											
Dark control / 2											
Dark control / 3											
Low light / 1											
Low light / 2											
Low light / 3											
Bright light / 1											
Bright light / 2											
Bright light / 3											

Source (virtual/classroom) for each treatment: _____

Trial summaries

Treatment	Trial 1	Trial 2	Trial 3	Mean
Dark control				
Low light				
Bright light				

Metric: ET50 in minutes. Use the first crossing of five disks; interpolate between readings. If not reached by 20 min, report that limit. Calculate a mean only when all three trials reach ET50.

Observations and graph

Graph time on the x-axis and the recorded measurement on the y-axis on separate graph paper, or use the app. Label treatments, units, and data sources. Record unusual observations below.

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3. Explain the evidence

ET50 is the interpolated time when 5 of 10 disks float. Lower ET50 suggests faster net oxygen accumulation under matched conditions. If five disks never float, report "not reached within 20 minutes"; do not treat that as a zero rate.

1. Which light condition had the shortest ET50? Support your claim with values from at least two treatments and discuss variation between trials.

2. Why are the dark control and matched leaf preparation necessary?

3. Explain how oxygen accumulation makes disks float and why this is an indirect measure of net activity, not a direct measurement of sugar production.

4. Identify one limitation or source of uncertainty and propose an improved investigation.

Before sharing: check units, source labels, calculations, and evidence. The app saves on this device only; download or print your report.

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Why infiltration matters

Air spaces inside a leaf can make a disk float before it produces oxygen. Infiltration replaces trapped air with solution, so all disks start sunken.

Put disks in bicarbonate solution in a needle-free syringe. Gently expel excess air, cover the tip, and pull back the plunger to reduce pressure. Trapped gas expands and escapes. Uncover the tip and release the vacuum gently so solution enters the spaces. Repeat as needed until disks sink; do not crush the leaves. The virtual model uses three pull/release cycles; real leaves vary.

Run a controlled comparison

In Collect observations, prepare the disks using Pull plunger and Release vacuum three times. Keep the assigned lamp setting: off (0%), low (25%), or high (100%). Start the timer and watch the lamp and disks. Use Pause, change timer speed, or advance two minutes. Record every two simulated minutes through 20 minutes. Use fresh disks for each independent trial.

Then experiment freely

Change the lamp during a run, or choose any intensity from 0 to 100%. After 20 minutes, resume to observe up to 120 minutes. Try high light, darkness, then high light again. Turning off the lamp does not pause time or instantly remove gas. Changed-light or incompletely infiltrated trials are labeled exploratory and excluded from comparison means. Download data before restarting a trial.

My change (setting and simulated time):

What happened immediately? What happened later? Explain using oxygen production, respiration, and trapped gas.

CSV and text reports include exploratory observations and lamp-change history. Model rates are illustrative, not predictions for real leaves.

Cut & sort: matter and energy

Optional introduction or extension | 10-15 minutes | Biology 2e, chapters 7 and 8.

Choose Card Activity on the lab overview to open the separate circular photosynthesis and respiration model, or arrange the printed cards on chart paper. Use two molecule sets for both halves at once, or reuse one 19-card set as you switch processes.

Print cutout pages in color, single-sided, at 100% scale. Cut along dashed lines. Use scissors safely. Group work: one person arranges, one counts atoms, and one explains; then rotate roles.

Model photosynthesis. Sort all the molecule cards into reactants and products. Add plus signs and the arrow, and decide where light and chlorophyll belong.

Count carbon, hydrogen, and oxygen atoms separately on each side. Explain how your arrangement shows conservation of matter and the transfer of energy. Energy symbols are annotations, not part of this simplified atom inventory.

This is an overall net equation, not a sequence of reaction stages. The glucose diagram shows an atom inventory, not a molecular structure.

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My equation and symbol placement:

Atom counts: Reactants C ____ H ____ O ____ | Products C ____ H ____ O ____

Explain where energy enters, is stored, or leaves this process:

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

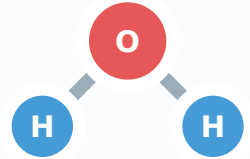
Build a balanced equation. Place light and chlorophyll where they help.



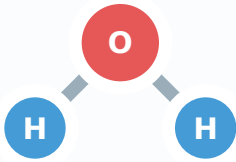
Oxygen



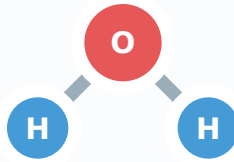
Carbon dioxide



Water



Water



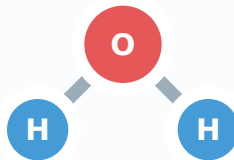
Water



Oxygen



Carbon dioxide



Water



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 1/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.

+



Combine with

O₂



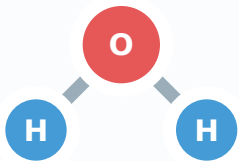
Oxygen

CO₂



Carbon dioxide

H₂O



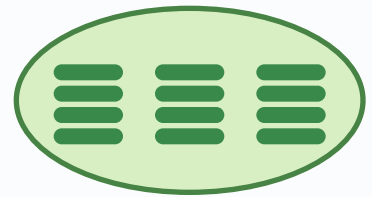
Water

+



Combine with

CHLOROPHYLL



Green pigment in chloroplasts
Simplified chloroplast diagram
Light-absorbing pigment

C₆H₁₂O₆



Atom inventory (not structure)

Glucose

O₂



Oxygen

LIGHT ENERGY



Sunlight

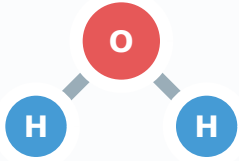
Atom colors: C = gray • O = red • H = blue

Student cutouts 2/3 • Print in color, single-sided, at 100%.

PHOTOSYNTHESIS • CUT & SORT

Cut on dashed lines. Keep all three sheets together.

Build a balanced equation. Place light and chlorophyll where they help.



Water



Reaction arrow



Oxygen



Oxygen



Carbon dioxide



Carbon dioxide

Atom colors: C = gray • O = red • H = blue

Student cutouts 3/3 • Print in color, single-sided, at 100%.